

Project Report

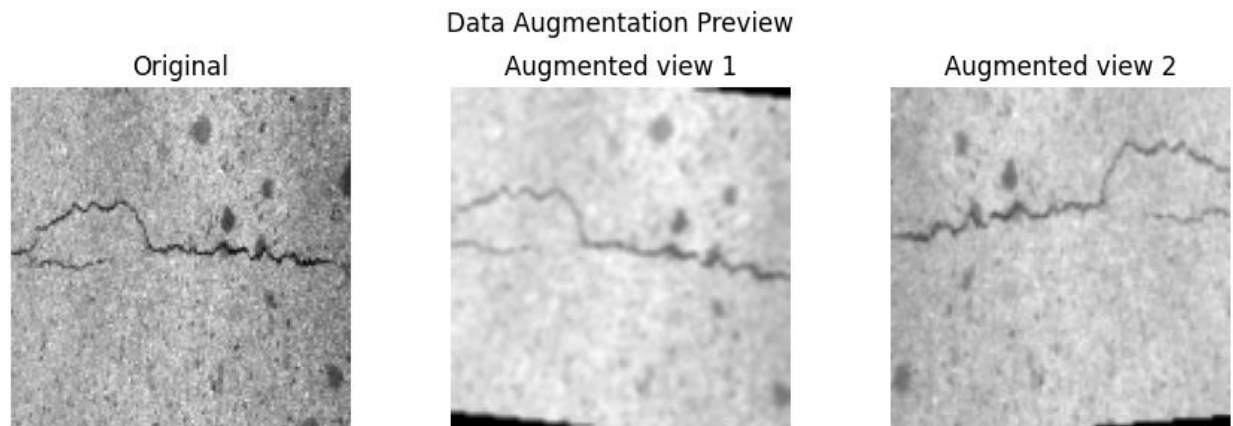
DSA4050: Unsupervised Learning in Computer Vision

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1. Introduction

This report presents the methodology, results, and analysis from a computer vision assignment applying unsupervised learning techniques to a road crack image dataset. The assignment covers five core tasks, image preprocessing, clustering, dimensionality reduction, contrastive learning, and comparative evaluation, as well as a bonus task implementing GAN-based feature learning. The central question addressed is: which unsupervised representation learning method most effectively captures the visual distinction between cracked and uncracked road surfaces without access to ground-truth labels during training?

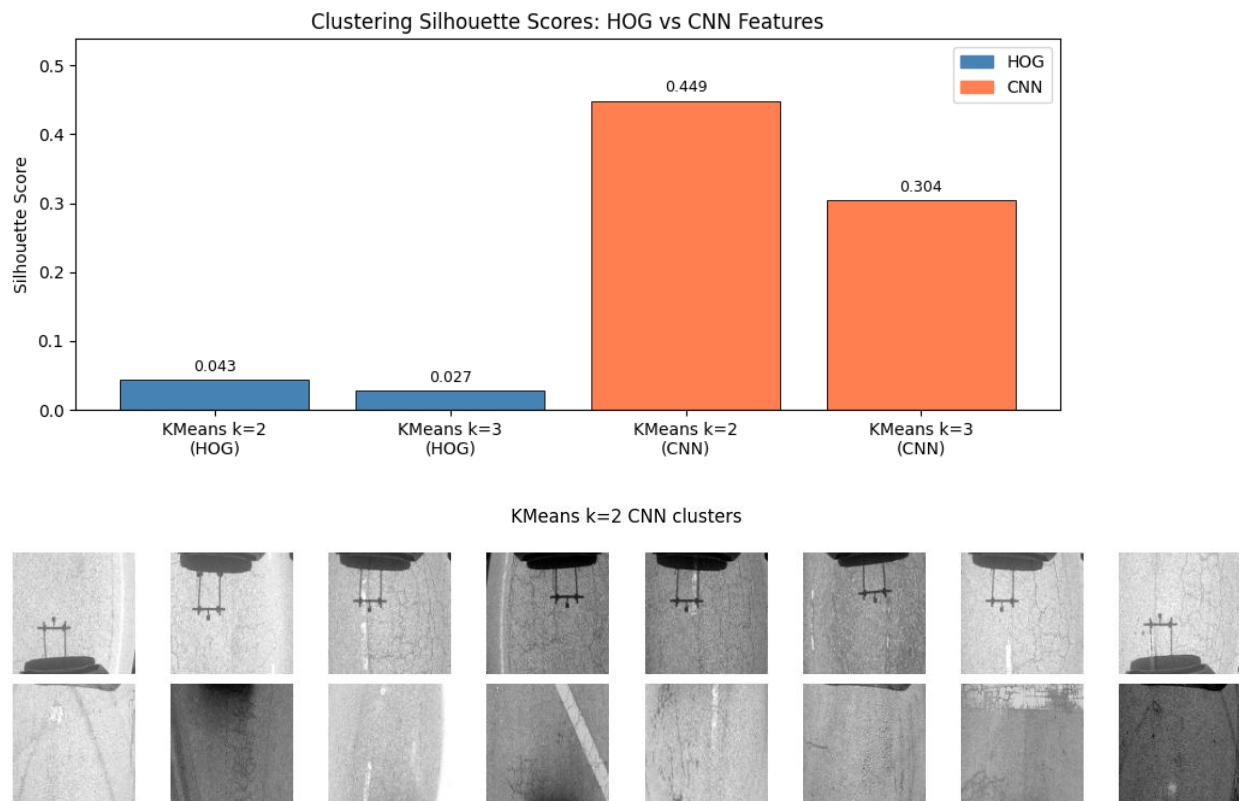
2. Dataset and Preprocessing



Two image collections were used. The first, `road_cracks`, contains 125 images of cracked road surfaces exclusively. The second, `train`, contains 1,389 images of both cracked and uncracked surfaces. The train dataset was stored in a biased order — uncracked images first, cracked images later — which was corrected by shuffling the file list in memory before loading, ensuring any subset drawn is representative of the full distribution.

All images were preprocessed using the following pipeline: conversion to grayscale (single-channel intensity), resizing to 128×128 pixels using area interpolation, and normalisation of pixel values to the range [0, 1] by dividing by 255. Grayscale conversion was chosen because crack patterns are fundamentally textural and structural, and single-channel representation halves memory requirements without meaningful information loss. A data augmentation pipeline was also defined — comprising random horizontal and vertical flips, rotations up to 30 degrees, random resized crops, colour jitter on brightness and contrast, and Gaussian blur — used exclusively during SimCLR training to generate positive pairs for contrastive learning.

3. Task 2: Clustering for Image Categorisation



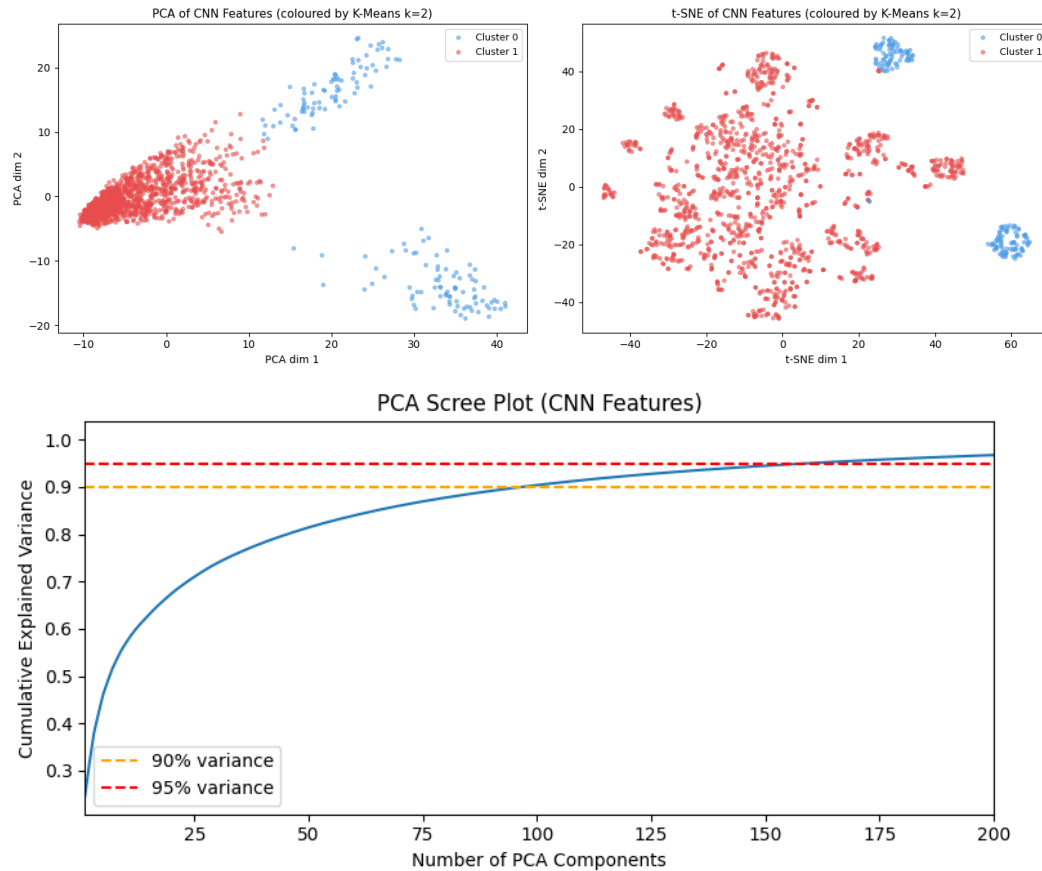
Feature extraction was performed using two strategies. Histogram of Oriented Gradients (HOG) produced a 1,764-dimensional descriptor per image by computing gradient magnitude histograms across 16×16 pixel cells with 9 orientation bins. A pretrained ResNet-18 backbone (ImageNet weights, classification head removed) produced a 512-dimensional deep embedding per image from its global average pooling layer. Both feature matrices were standardised before clustering.

K-Means was applied with $k=2$, 3, and 4, using PCA pre-reduction to 64 components. DBSCAN was tested across epsilon values of 0.5, 1.0, and 2.0 with minimum samples of 5. Cluster quality was evaluated using the Silhouette Score, which ranges from -1 (incorrect clustering) to $+1$ (dense, well-separated clusters).

DBSCAN failed to produce any valid clusters under all tested configurations for both feature types, returning all images as a single cluster or noise. This is consistent with the continuous, unimodal nature of road surface feature distributions — the data lacks the isolated high-density pockets that DBSCAN requires.

K-Means results revealed a dramatic difference between feature types. HOG features produced silhouette scores of 0.043 ($k=2$), 0.027 ($k=3$), and 0.042 ($k=4$) — values close to zero indicating near-random cluster assignments with no meaningful separation. CNN features produced silhouette scores of 0.449 ($k=2$), 0.304 ($k=3$), and 0.270 ($k=4$). The best result, CNN $k=2$ with a score of 0.449, represents a moderately strong clustering result and aligns with the binary nature of the dataset. Visual inspection of the $k=2$ CNN clusters confirmed that the separation is meaningful — one cluster predominantly contains cracked surfaces and the other uncracked surfaces, though some boundary mixing is present. HOG clusters showed no meaningful visual coherence. The CNN feature advantage is more than tenfold in silhouette score, confirming that deep semantic features are essential for this task.

4. Task 3: Dimensionality Reduction and Visualisation



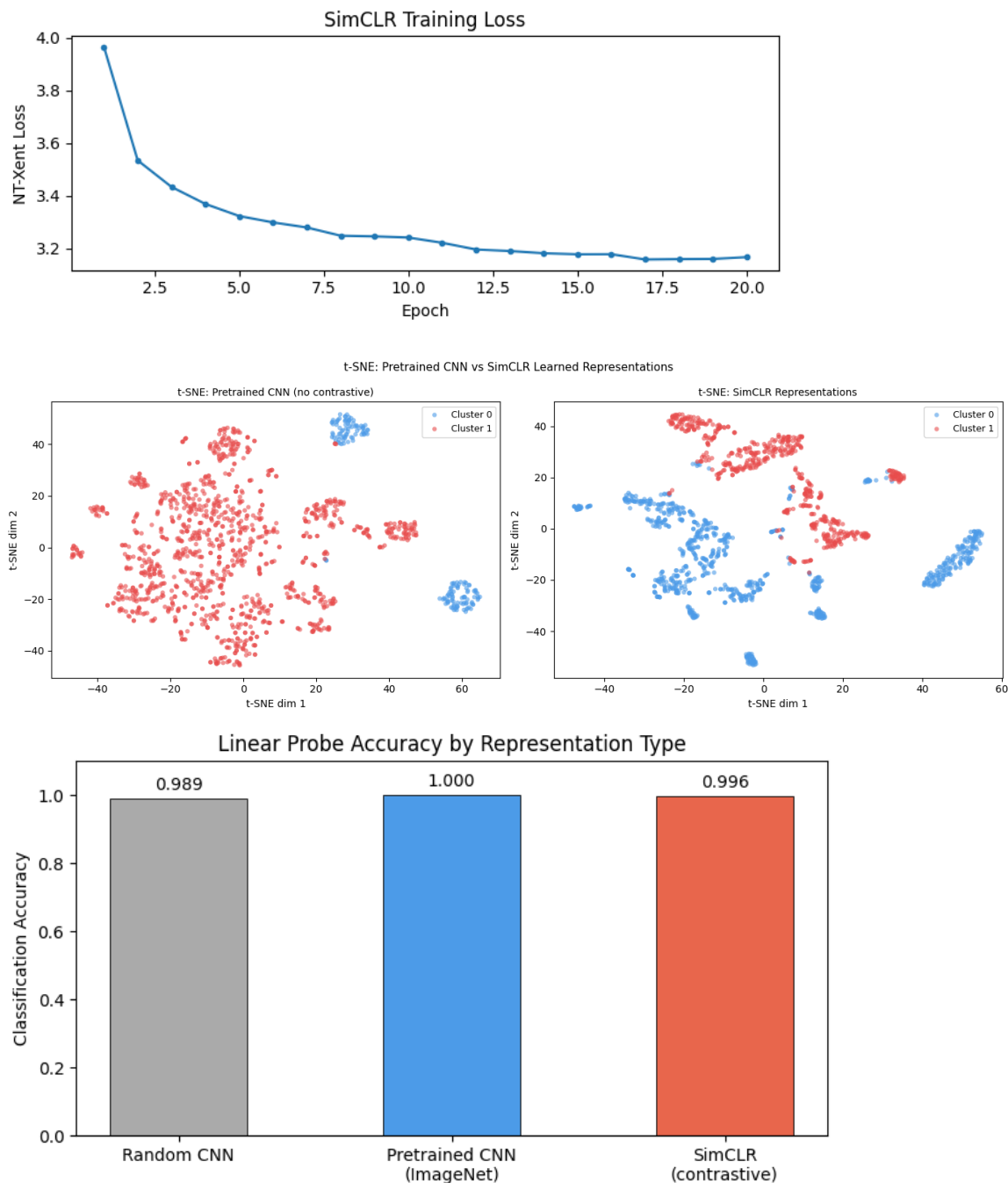
PCA and t-SNE were applied to the 512-dimensional CNN features to enable 2D visualisation of the representation space.

PCA reduction to 2 components captured PC1 = 24.5% and PC2 = 7.1% of total variance, for a combined 31.6%. This relatively low figure reflects the distributed nature of deep representations — no single linear direction dominates the variance. The scree plot confirmed this: the cumulative variance curve follows a logarithmic shape, requiring 97 components to capture 90% of variance and 160 components to capture 95%. In the 2D PCA scatter plot, Cluster 1 formed a compact grouping near the origin (x: -10 to 10, y: -10 to 10). Cluster 0 was more dispersed, splitting into two sub-groups: one in the lower-right region (x: 20 to 40, y: -20 to -10) and one in the upper-right (x: 10 to 30, y: 10 to 20). This bimodal spread within Cluster 0 suggests two visually distinct sub-types, possibly different crack severities or surface materials.

t-SNE, applied after PCA pre-reduction to 50 components, revealed a complementary view. Cluster 1 was spread through the centre of the embedding space rather than being compact, reflecting that t-SNE preserves local neighbourhoods rather than global variance — Cluster 1 images are locally diverse but do not form a single tight global manifold. Cluster 0 showed two clearly compressed, distinct groups: one in the top-right and one in

the bottom-right. These dense pockets strongly corroborate the bimodal structure observed in PCA. Together, PCA and t-SNE tell a consistent story: the dataset contains one broad cluster of uniform surfaces and a second cluster of two visually coherent crack sub-types.

5. Task 4: Contrastive Learning with SimCLR



SimCLR was implemented using a ResNet-18 encoder producing 512-dimensional embeddings, followed by a two-layer MLP projection head mapping to a 128-dimensional normalised space. The NT-Xent (normalised temperature-scaled cross-entropy) loss with temperature 0.5 was used as the contrastive objective, training the model to maximise cosine similarity between two differently-augmented views of the same image (positive pairs) while minimising it against all other images in the batch (negative pairs). Training used the combined pool of all 1,514 images (road_cracks + train) for 20 epochs with the Adam optimiser and cosine annealing scheduler.

The training loss decreased from 3.965 at epoch 1 to 3.167 at epoch 20, a 20% reduction, with a smooth, consistent descent and no signs of instability or divergence. The loss did not fully converge, indicating that additional epochs (50–100 on GPU) would yield stronger representations, particularly important given the small dataset size limiting the number of negative pairs per batch.

The t-SNE comparison between pretrained CNN and SimCLR representations showed a structural reorganisation of the embedding space. In the pretrained CNN plot, Cluster 0 formed two dense compressed groups at the top-right and bottom-right. In the SimCLR plot, Cluster 0 formed a more moderately spread group on the middle-right while Cluster 1 migrated toward the upper-centre. The SimCLR embedding reflects a different notion of similarity — images that look alike under photometric and geometric distortions — rather than ImageNet category proximity.

Linear probe classification used logistic regression trained on an 80/20 train/test split using proxy labels derived from K-Means k=2 CNN clustering. Results across all three representation types were high:

- Random CNN (no pretraining): 98.92%
- Pretrained CNN (ImageNet): 100.00%
- SimCLR (contrastive, 20 epochs): 99.64%

The near-perfect random baseline is notable and reflects that the convolutional architecture's inductive bias — local filters, spatial pooling — is itself well-matched to distinguishing road surface types, regardless of trained weights. SimCLR's 99.64%, achieved with no labels and no ImageNet data, demonstrates that contrastive pretraining successfully learns discriminative road-specific features within 20 epochs. Both SimCLR and the pretrained CNN misclassify at most one image from the minority class (Class 0: 32 test samples), with the class imbalance (162 cracked vs. 1,227 uncracked) being the primary source of residual error.

6. Task 5: Comparative Analysis



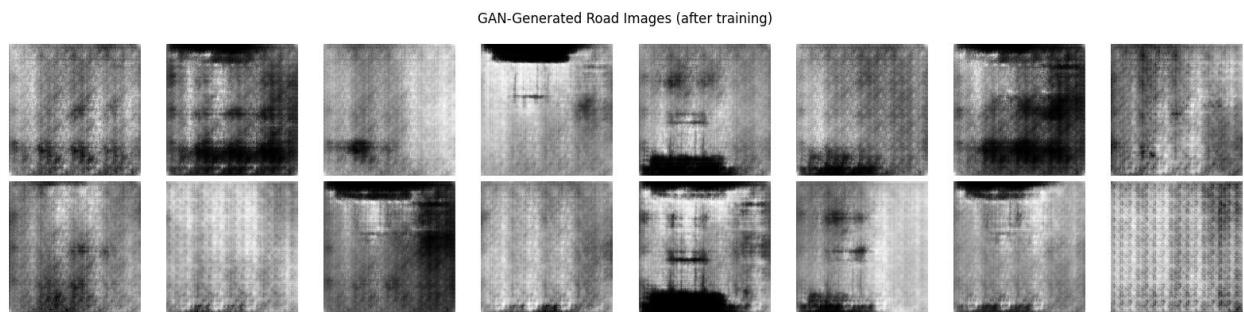
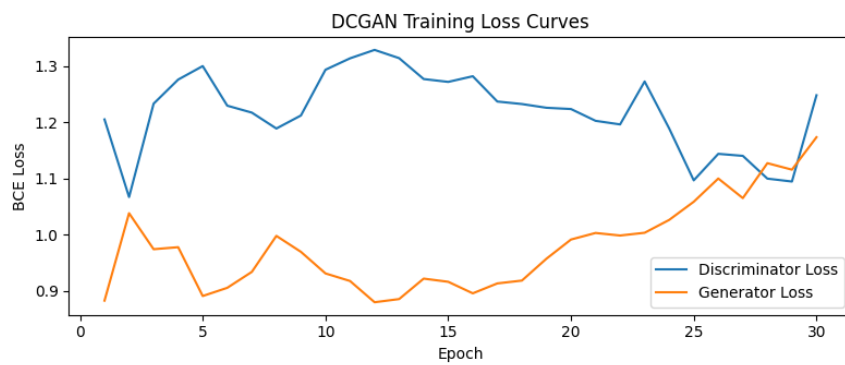
Method	Features	Silhouette Score	Classifier Accuracy
K-Means k=2	HOG	0.043	—
K-Means k=3	HOG	0.027	—
K-Means k=2	CNN	0.449	1.000
K-Means k=3	CNN	0.304	—
K-Means k=2	SimCLR	0.123	0.996

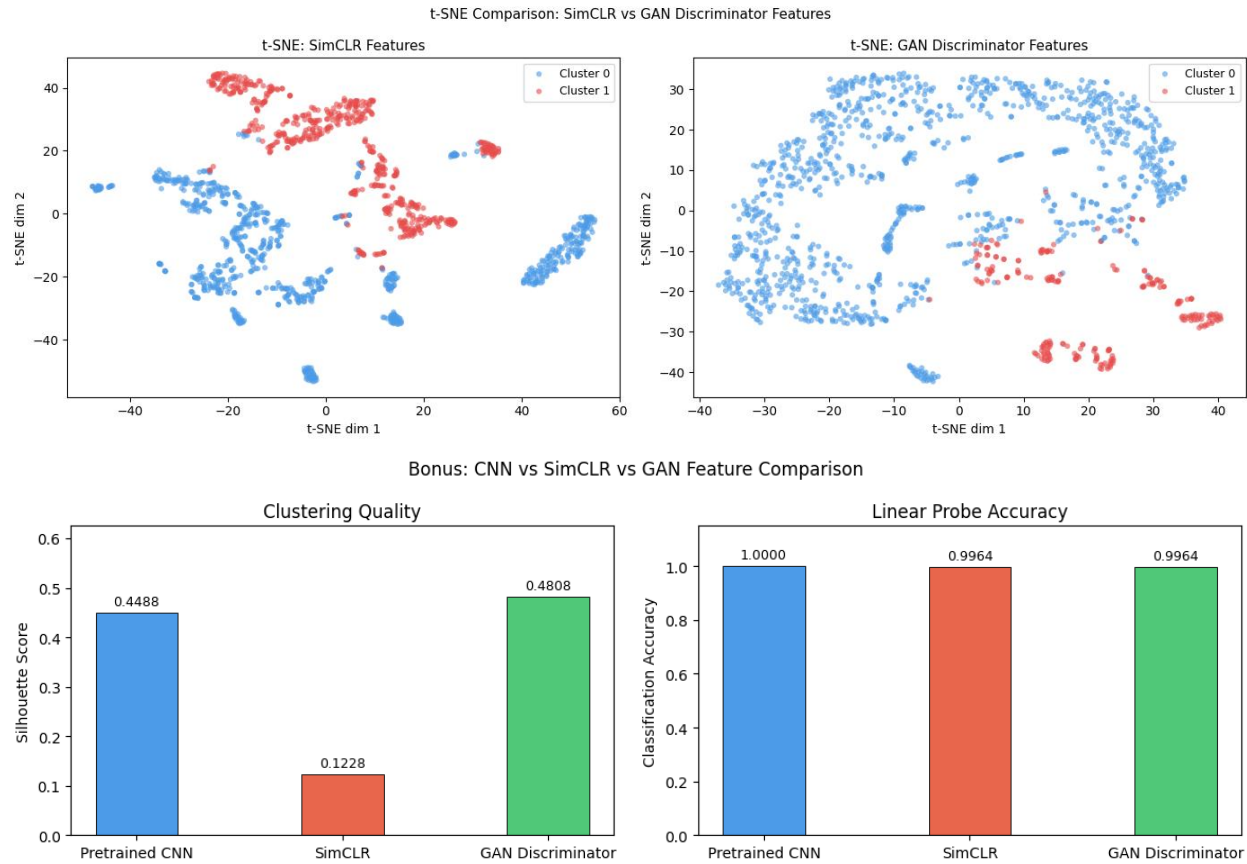
The results establish a clear hierarchy across all methods. Feature quality is the dominant factor in both clustering and classification performance — the gap between HOG and CNN silhouette scores (0.043 vs 0.449) is larger than the gap between any two deep feature methods. DBSCAN was ineffective across all feature types, confirming that density-based clustering is poorly suited to the continuous, high-dimensional distribution of road image features.

Among deep feature methods, the pretrained CNN achieves perfect classification but its clustering superiority over SimCLR (0.449 vs 0.123) reflects the structural difference in their objectives: ImageNet pretraining naturally separates visually distinct categories, while SimCLR's augmentation invariance objective organises the space differently. SimCLR's

lower clustering score does not indicate inferior representations — it indicates a different geometry that is highly discriminative for linear classification but less amenable to K-Means partitioning.

7. Bonus Task: GAN-Based Feature Learning





A DCGAN was trained from scratch on the 1,389-image training set for 30 epochs. The generator (3.6M parameters) maps a 100-dimensional latent vector to a 128×128 grayscale road image through five transposed convolutional layers. The discriminator (2.8M parameters) maps images to a real/fake probability through five strided convolutional layers, with a 512-dimensional global average pooled feature extracted from the penultimate layer as the unsupervised representation.

Training was stable throughout all 30 epochs. The discriminator loss fluctuated between 1.093 and 1.299 while the generator loss rose from 0.883 to 1.173 — a pattern indicating the generator became progressively harder to fool as the discriminator refined its internal model of real road surfaces. Neither loss spiked nor collapsed, and both moved toward the BCE equilibrium value of $\ln(2) \approx 0.693$, confirming that no mode collapse occurred. The generator had not yet reached equilibrium at epoch 30, suggesting further training would improve feature quality.

The final comparison across all three deep feature methods is as follows:

Method	Silhouette Score	Classifier Accuracy
Pretrained CNN (ImageNet)	0.4488	1.0000
SimCLR (contrastive)	0.1228	0.9964
GAN Discriminator	0.4808	0.9964

The GAN discriminator achieved the highest silhouette score of all methods (0.4808), surpassing even the ImageNet-pretrained CNN (0.4488) and substantially outperforming SimCLR (0.1228). It matched SimCLR exactly on classifier accuracy (99.64%), with both methods correctly classifying 277 of 278 test images and misclassifying the same single sample from the minority cracked class.

The GAN's clustering superiority is explained by the adversarial training dynamic: the discriminator is continuously pressured to find finer-grained features that distinguish real road surfaces from increasingly convincing fakes. This process, trained entirely on domain-specific road images, shapes the feature space around the natural modes of variation in road surfaces, including crack presence, more directly than general ImageNet pretraining. SimCLR's lower clustering score reflects its design choice to optimise for augmentation invariance rather than cluster separation; its representations are organised along different axes that support classification but not K-Means partitioning as strongly.

8. Conclusions

This project demonstrated that unsupervised representation learning can achieve near-perfect categorisation of road crack images without any label supervision during training. The central findings are:

Feature quality is the single most important factor. HOG features are insufficient for this task, their silhouette scores of 0.027–0.043 indicate the feature space provides almost no cluster structure. Deep convolutional features, whether pretrained, contrastively trained, or adversarially trained, all produce dramatically superior representations with classifier accuracies above 98.9%.

Among unsupervised methods, the GAN discriminator produced the best cluster structure (silhouette 0.481), while SimCLR and the GAN matched on downstream classification accuracy (99.64%). The pretrained CNN achieved perfect classification but required large-scale supervised pretraining on an unrelated dataset (ImageNet), making it less applicable in scenarios where no such pretrained model is available.

DBSCAN was ineffective for this dataset and should not be recommended for image clustering tasks without first applying aggressive dimensionality reduction or carefully tuning epsilon through k-distance graph analysis.

The main limitation across all methods is dataset size. With approximately 1,400 images and a significant class imbalance (88% uncracked, 12% cracked), both SimCLR and the GAN are operating below their intended data regimes. SimCLR's contrastive objective benefits from large batch sizes to provide sufficient negative pairs, and the GAN's generator quality would improve substantially with more training data and epochs. Future work should collect a larger, more balanced road surface dataset and evaluate whether the GAN's clustering advantage over SimCLR persists at scale, where SimCLR's architectural simplicity and principled objective may give it a long-run advantage.